

A Gamified Citizen Approach for the Collection of Data to Train Deep Learning Models

Chantelle Saliba

Abstract—Machine Learning has been a fundamental stepping stone in the era of Artificial Intelligence. It has been proved by a number of researchers that Machine Learning has been an effective solution in numerous fields such as in the medical field, in psychology and image classification. In this research, we will focus on the limits of Machine Learning by attempting to build a model that makes use of limited training data. The case scenario that will be analysed is that of the Maltese flora that in itself does not contain sufficient training material. Therefore, citizen science is going to be used to collect the required data to train the model. In this research, we will see how an android application was developed for data collection and an evaluation strategy that shows the effect of citizen science on Machine Learning was devised and presented. In addition, different Machine Learning models were evaluated. In the coming weeks, the android application will be developed further and training on the first dataset will be performed.

Index Terms - machine learning; deep learning; data collection; gamification; citizen science; neural networks; object recognition; Maltese flora; botany;

I. INTRODUCTION

A. Problem Definition

Machine Learning (ML) is one of the fundamental concepts of Artificial Intelligence and it is playing a key role in the advancement of technology. One can find ML in various fields such as in robotics, computer games, data mining, medical diagnosis, social media services, pattern recognition, natural language processing, product recommendation, online fraud detection and many more [1]. For these fields, however, numerous data is available and therefore building and training deep learning models is possible and feasible [2-11]. But what happens in scenarios when there is not sufficient training data? Should such fields be disregarded and therefore lack technological advancements?

Training a deep learning model that makes use of minimal data or rather that is continuously gathering data to train itself is important especially to scenarios where there are lack of resources, making data sparse. An example of such scenario is the classification of Maltese flora. Flora is defined as plants and flowers that one finds in a particular region at a particular time. In Malta, due to our central position in the Mediterranean sea, one finds flora that is located in southern Europe and northern Africa as well as flora that is indigenous to the Maltese islands [12, 13]. To date, there is not sufficient digital resources that will allow the training of a deep learning model in the conventional way.

B. Motivation

The motivation of this project is divided into two main concepts. The first concept is from a technological point of view and the second concept is from a natural point of view.

The introduction of ML in less known fields such as botany and in fields that contain sparse data is something that should be considered for the improvement of this technological era. ML has been introduced to many different fields and it has been seen to be effective and helpful in everyday life but now it is time to expand the uses of ML and test its limits.

As Sir David Attenborough once said “Cherish the natural world because you’re part of it and you depend on it”. Many of us are not informed and appreciative about our nation’s flora while some tourists visit Malta specifically to come and experience Malta’s nature and what it has to offer. This project will also be a good and educational tool for local hikers. This project will also be beneficial to botanists because apart from the instant recognition of species, information regarding the geographic spread of the flora can be retrieved from the model.

C. Why the Problem is non-trivial

ML has seen drastic improvements throughout the years and models have been built in a numerous numbers of fields. However, not many studies have been conducted on how ML can cater for models with limited data.

II. AIMS AND OBJECTIVES

The aim of this project is to develop a ML model that makes use of limited training data by making use of gamification and citizen science. As a data source, with the assistance of a local academic botanist, the Maltese flora is being considered.

To achieve this aim the following objectives have been set:

- Devise a method to gather data through gamification and citizen science.
- Design and implement a user centric mobile application that will enable interaction between the end-user and the available data to enrich the existing dataset.
- Implementation of a deep learning model that will be able to classify Maltese flora and gather statistical information making use of a data collection system.

III. BACKGROUND RESEARCH AND LITERATURE REVIEW

A number of works on flower image retrieval, classification and gamification have been carried out throughout the years. Flora is one of the most important elements of life for a region

making life itself possible. Therefore, numerous researchers have sought to make flower classification feasible. Gamification is the process of adopting game design mechanics to modern business practices outside the gaming industry. Gamification helps engage and motivate its users through a number of game elements such as rewards, points, characters, feedback and fun. [14, 15]

A. Flower Recognition Systems

Image feature recognition is used in a wide area of applications and a lot of research has been put in that particular field of computer vision. Extracting the flower region from the image with high accuracy has proven to be a challenge due to complex backgrounds. Since the images are taken in a real environment, illumination conditions may vary according to the weather and time of the day. Background features can also become more prominent overshadowing the flower region. In addition, variations in viewpoint and scale can also have an impact on the region extraction.

Liu *et al.* in [16] proposed a system in which the saliency map and the luminance map of an image are combined. A saliency map is a topographical representation of the unique features of the image while a luminance map shows the levels of light in an image. The bottom-up saliency map is calculated by using a regional contrast based salient object detection algorithm. This system makes use of the prior knowledge that flowers have high brightness.

Pei and Cao in [17] implemented the Grab Cut algorithm [18] to segment the flower region from the background. This algorithm makes use of user input and iteratively the method will aim to assign each pixel in the image a label; background or foreground. This system then finds the coordinates of the flower region centre and with this data, two feature-rings are extracted. The first ring contains pixels with a distance within the interval $(\frac{5}{6}r, r)$ to the centre. From this ring, information about the main colour components and the number of connected components can be retrieved. The second ring is made up of pixels with a distance within the interval $(\frac{1}{3}r, \frac{2}{3}r)$ to the centre. Pei and Cao altered the Gray Level Co-Occurrence Matrix [1] method to be performed in the polar coordinate system, retrieving statistic values such as contrast, correlation, energy and homogeneity.

A similar approach was taken by Tiay *et al.* in [19]. Here the Graph-Cut algorithm [20] was used to eliminate the background of the image. The Graph-Cut algorithm is built on the Grab-Cut algorithm [18]. The approach in [19] follows by transforming the image to greyscale and removing any surplus background by searching for the top, bottom, right and left extents of the flower. Canny Edge Detection algorithm is then being used to receive edge data. This data is then being used in Hu's Seven Moment Algorithm to obtain seven characteristics of the flower image.

A completely different approach was taken by Zawbaa *et al.* [21] where the image is transformed to the LAB colour

space [2] and the frame which includes the flower is selected by the user. Delta E, which is the measure of change in visual perception, is then used to calculate the distance between every pixel in the selected frame and the average LAB colour. Otsu's threshold is then applied on each LAB colour to segment the flower. Otsu's threshold algorithm is a binarization algorithm whose aim is to find the value where the sum of the foreground and background spread is at its minimum [22]. This paper proposes two different techniques for feature extraction; Scale Invariant Feature Transform (SIFT) and Segmentation-based Fractal Texture Analysis (SFTA). SIFT [23] is an algorithm that is used to detect certain key points within an image. This algorithm is particular useful in this use case as it is invariant to image translations, scaling, rotation and partially invariant to illumination changes and affine projection. SFTA [24] is an algorithm that makes use of multi-thresholding on an image to decompose the segmented image into several components. SFTA is made up of two main parts. The first part is binary decomposition and the second part is feature extraction.

Hong and Choi in [25] proposed the use of colour-based contour detection and edge-based contour detection as methods of feature extraction. The system starts by acquiring a smoother image and this is achieved through the use of a Median Filter [3]. Dilation and erosion operations [4] are then performed. This is followed by changing the background into a black background to enable contrast with the colours of the flowers. For this paper, Hong and Choi have chosen five main flower colours: yellow, red, pink, white and blue. The image will then be passed through colour-based contour detection. The proposed system will compare the hue value of the reference pixels to nearby pixels since the background pixels are similar in colour. If the difference is less than five, the pixel is considered to be part of the flower, while if the difference is greater than five, the process is stopped and the last pixel is considered to be the contour of the flower. The image will then be passed through Canny Edge Detection. During this phase, the edge of the flower is detected by finding a point in which the image intensity undergoes a rapid change.

B. Flower Classification Systems

Flower classification is the basic of botanical studies and it has been around since the 18th century [16] where Carl Linnaeus proposed a level hierarchical plant classification system that nowadays can recognize up to 369,000 species. This type of classification, however, requires expert and domain-specific knowledge. Therefore, such a system is not widely accessible since the majority of the people do not have the required knowledge.

Researchers have been trying to automate flower classification systems throughout the years in order to make them

²More information available at "CIELab Color Space" by Gernot Hoffmann (2019)

³More information available at Chapter 4 of "Nonlinear Digital Filters" by Pitas I. and Venetsanopoulos A.N (1990)

⁴More information available at "Image Processing and Mathematical Morphology: Fundamentals and Applications" by Frank Y. Shih (2010)

¹More information available at "Textural Features for Image Classification" by Robert M Haralick, K Shanmugam and Its'hak Dinstein (1973)

accessible and with highly accuracy results. [16, 19, 21, 25–28] In this section, a review of different approaches to the problem of flower classification will be evaluated.

1) *Machine Learning*: ML is an Artificial Intelligence application that allows a system to learn and improve with experience. ML systems build a mathematical model based on the training data and from this data the system will be able to make predication on data that it never saw before. ML is being applied to various fields such as psychology [2], medical diagnosis [3] and image classification [19, 21, 25], [4].

Tiay *et al.* in [19] made use of the K-Nearest Neighbour Algorithm [1]. This algorithm is a supervised ML technique in which it takes a set of labelled data points grouped into several classes and tries to classify new given data points into their respective class. The algorithm works by calculating the distance between that data point and the other training points making use of a similarity function such as the Euclidean, Jaccard or Cosine distance. The algorithm will take the closest K number of classes and computes a majority vote. The data point will be classified within the class that had the majority of votes [29]. Amongst other use cases, K-Nearest Neighbour Algorithm can be used for medical condition classification [5] and for phoneme prediction [6]. As input, the researchers in [19] made use of the characteristics retrieved from the edge and the colour of the image that were extracted during the feature extraction. The system gave an accuracy of over 80%.

Zawbaa *et al.* [21] made use of two ML techniques; Random Forests (RF) and Support Vector Machines (SVM). Zawbaa *et al.* compared the results generated by both techniques. The RF algorithm [8] is a supervised ML method, used for both classification and regression. RFs [7] are made up of a collection of individual decision trees [1]. Each individual tree will give out a class prediction. The class with the most votes in the RF will become the model's prediction. Some other examples where a RF algorithm was implemented as a ML technique are in disease prediction [7] and credit card fraud detection [8]. The SVM [1] is a supervised ML technique in which given training data, the algorithm outputs an optimal hyperplane which is the decision boundary. This will be used to categorize new examples. Some fields in which SVMs were used are in the electrical field [9], medical research [10] and agriculture [30]. As input [21] is taking the feature vectors extracted from each image either from the Scale Invariant Feature Transform (SIFT) or from the Segmentation-based Fractal Texture Analysis (SFTA) algorithms.

Hong and Choi in [25] made use of more than one technique in order to perform flower classification. In this paper, Hong and Choi made use of colour clustering and distance histogram matching in order to improve the recognition rate. Moreover, the authors also proposed partial matching. Colour clustering was proposed through the use of K-means clustering. The authors made use of two-dimensional clustering by making use of the hue and the saturation values. K-means clustering [1, 31] is an unsupervised, iterative ML algorithm which is required to classify data into k different clusters converging to a local minimum. Each cluster represents a class. For every

given data point the Euclidean distance between that point and all the centres of each cluster are calculated. The inputted data point is said to belong to the class with the minimum centre distance. The locations of the centroids will be recalculated until these are optimised. The second proposition for classification is distance histogram matching [32]. The proposed approach was the history matching technique in which the distance values in the histogram bin are compared to the histogram values of the reference images previously saved in the server. The crossover method was also proposed to measure the similarity of histogram values. Another technique proposed by Hong and Choi in [25] was partial matching. In partial matching the image is to be split into two directions vertically or horizontally so that a total of five types of images can be compared. The authors of this paper found that dividing the image by four has the highest recognition rate. The main motive behind partial recognition is to handle the problem of light. From this study, Hong and Choi concluded that applying both colour-based and edge-based contour detection generates higher recognition rates than when performing colour-based detection or edge-based detection alone. The system gave an accuracy of 85.2% for the first rank, 89.8% for the second rank and 94.8% for the third rank.

2) *Convolutional Neural Network*: Convolutional Neural Network (CNN) [11, 16, 33, 34] is a deep learning algorithm which is biologically inspired. It receives an image as input and passes it through a series of convolutional layers followed by one or more fully connected layers. A CNN is made up of two components; Feature Learning and Classification. In the feature learning component, the CNN learns about the training data by passing each image through the convolutional layer and the average pooling layer for a number of times determined by the number of layers in the CNN. The convolutional layer is used to learn the image features by computing convolutions between the input and a set of kernels or filters. A kernel or filter can perform operations such as edge detection and sharpening. A non-linear activation function such as the rectified linear unit (ReLU) or the softplus function is then performed on the convolutions. This introduces non-linearity in the convolutional network. The pooling layer tries to reduce the number of parameters when the images are too large and reduces the dimensionality of each map but retains important information. After feature extraction, the classification component receives the data as a matrix and after performing flattening, passes it through one or more fully connected layers. The results will then be passed through an activation function such as softmax or sigmoid to classify the outputs. CNNs can be seen at work in various areas such as music classification [33], hand gesture recognition [34] and classification of chronic myeloid leukemia [11].

Liu *et al.* in [16] made use of an eight layer CNN to classify flower images. The first five layers were convolutional layers while the remaining three were fully connected layers. The first and second convolutional layers had 64 and 128 kernels respectively, while the third layer had 256 kernels. The fourth and the fifth convolutional layers had 512 kernels each. The

first and second fully-connected components had 1024 and 512 channels respectively while the last fully connected component is a 79-way classification which is making use of the softmax activation function. This will produce a distribution over the 79 class labels. Some methods that were introduced in the CNN were local response normalization, overlapping pooling and dropout. Local response normalization prevents saturation. This was applied after the ReLU non-linearity in certain layers. Overlapping pooling controls overfitting in the pooling layers while dropout was used to avoid substantial overfitting in the fully-connected layers. The proposed system achieves a 76.54% success rate.

3) *Transfer Learning*: Transfer Learning [35, 36] is a ML method in which we can use existing knowledge learnt from solving one problem to solve other different but related problems.

Xia *et al.* in [26] experimented with transfer learning. The paper here used the Inception-V3 model [37] which is a pre-trained model on TensorFlow⁵. This deep neural network has been trained on more than a million images from the ImageNet database. To implement such a model for this use case, the authors kept the parameters of the previous layer and removed the last layer of the model. The flower data set was then inputted to retain the last new layer. The number of output nodes in the last layer was made equal to the number of categories in the dataset. The last layer of the model was trained by the back propagation algorithm and the cross entropy cost function was used to adjust the weight parameter. This was achieved by calculating the error between the output of the softmax layer and the label vector of the given sample category. This proposed system generated an accuracy of a 94% to 95%.

Another model that is used for transfer learning is the MobileNet Model as shown in [27]. The architecture of the model is more suitable for mobile and embedded based vision applications where there is lack of compute power. MobileNet [38] is built on depth wise distinguishable filters which uses depth wise intricacy that applies a distinct filter to each input channel. The depthwise distinguishable complexity separates into two layers; filtering and combining. This allows for reduction in computational time and size of the model. To train the model on the flower case, the authors of the paper did not change the lower layers of the network but computed the layers behind the layer just before the final output to fit the new data set. The proposed system generated an accuracy of between 85% and 99%.

Wu *et al.* in [28] discussed and compared the VGG16, VGG19, Resnet50 and Inception-v3 models. All these four models are pre-trained models that were trained on the ImageNet dataset which contains more than a million images. The VGG16 model and the VGG19 model [39] are two deep convolutional networks used for object recognition, made of 16 and 19 layers respectively. The ResNet50 [40] is a convo-

TABLE I: Comparison of overall performance of different transfer learning methods when classifying flowers

Method	Overall Performance
VGG16	83.53% - 83.82%
VGG19	84.71% - 85.25%
Inception-V3	94.58% - 95.88%
ResNet50	95.29% - 96.57%

TABLE II: Evaluation of Machine Learning Methods and their Accuracy on their own Flower Datasets

Reference	Image Classification Method	Accuracy
[25]	K-Means Clustering, Distance Histogram, Matching and Partial Matching	1st Rank - 85.2% 2nd Rank - 89.8% 3rd Rank - 94.8%
[19]	K-Nearest Neighbour Algorithm	80%+
[21]	SIFT with Support Vector Machine (SVM)	~72% to 100%
[21]	SIFT with Random Forest (RF)	~66.67% to 100%
[21]	SFTA with Random Forest (RF)	~77.78% to 100%
[21]	SFTA with Random Forest (RF)	~80% to 100%

lutional network that is 50 layers deep. The authors in order to perform transfer learning transformed the model parameters to the flower data set. This was achieved by transferring the weights of the network-specific layers in each model and fine-tune the model by training and learning the original data. The overall performance achieved by transfer learning is shown in table I. One can analyse that the Inception-V3 model and the ResNet50 model had higher accuracy rates than any other method.

4) *Results*: In this section, we present an overview of all the systems that were proposed and their generated accuracy rates. Different papers made use of different datasets. Some datasets were created by the authors themselves while some authors opted to make use of the Oxford-17 [41] or the Oxford-102 [42] flower dataset. In Tables 2, 3 and 4 one can analyse the different approaches that different researchers took to solving the flower recognition problem.

IV. DATASETS

The most common datasets when it comes to flower classification are the Oxford-102 and the Oxford-17 datasets. The Oxford-102 [42] and the Oxford-17 dataset [41] were created by Nilsback and Zisserman and they consist of flowers

TABLE III: Evaluation of a Convolution Neural Network and its Accuracy on Flower Datasets

Reference	Image Classification Method	Dataset	Accuracy
[16]	Convolution Neural Network with 8 layers	Oxford-102	84.02%
[16]	Convolution Neural Network with 8 layers	Own Dataset	76.54%

⁵Open source machine learning framework by Google available at <https://www.tensorflow.org/>

TABLE IV: Evaluation of Transfer Learning Methods and their Accuracy on Flower Datasets

Reference	Transfer Learning Classification Method	Dataset	Accuracy
[26]	Inception-v3	Oxford-17	95%
[26]	Inception-v3	Oxford-102	94%
[27]	MobileNet	Own Dataset	85% to 99%
[28]	VGG16	Oxford-17	98% to 99%
[28]	VGG16	Oxford-102	91% to 92%
[28]	VGG19	Oxford-17	98% to 99%
[28]	VGG19	Oxford-102	92% to 93%
[28]	Inception-v3	Oxford-17	99% to 100%
[28]	Inception-v3	Oxford-102	99% to 100%
[28]	ResNet50	Oxford-17	99% to 100%
[28]	ResNet50	Oxford-102	99% to 100%

commonly found in the United Kingdom. The Oxford-102 is made up of 102 categories while the Oxford-17 is made up of 17 categories.

Another less known dataset is the Edible wild plants dataset [43]. This dataset contains images of 62 edible wild plants.

V. PROPOSED SOLUTION

The proposed solution will be based on CNNs as proposed by Liu *et al.* in [16] to perform classification. The proposed solution will be making use of a combination of datasets: Oxford-102 [42], Oxford-17 [41] and the Edible wild plants [43] dataset. The dataset will be fine-tuned to fit the criteria of Maltese flora and will be analysed and verified by the dissertation co-supervisor Prof. Buhagiar from the department of biology from the Faculty of Science.

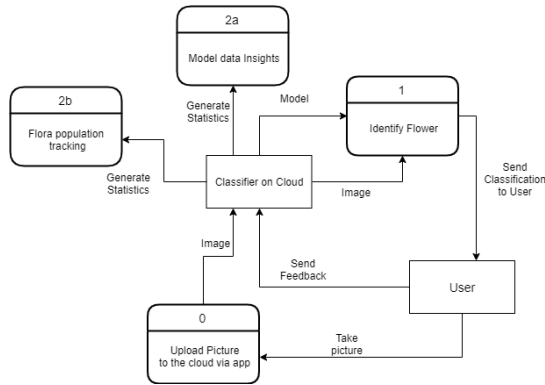


Fig. 1: Block Diagram of the Proposed Solution for the User

The proposed solution will be targeting two different types of people: the user with limited botany knowledge and a user with expertise in the field. Both users have different aims and these aims will be targetted and addressed.

Figure 1 shows the block diagram that will be dedicated to the user. The user does not have expertise in the field and therefore will not be able to contribute images and labels to the model. However, the user will be able to request for information and provide feedback to that model. As the flow diagram shows the user will be interacting with the model through the use of an application and will be sending the

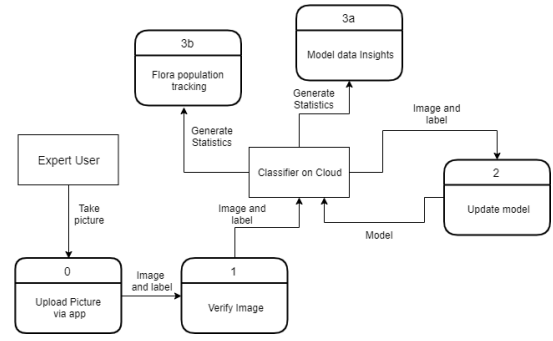
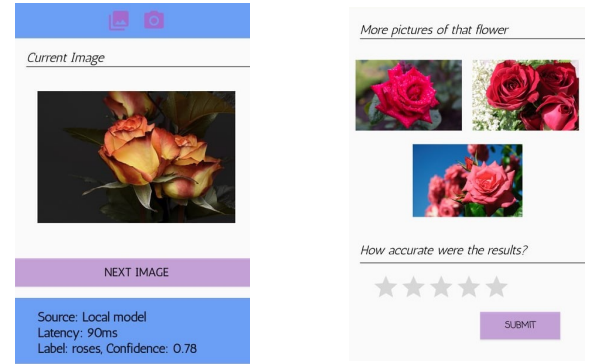


Fig. 2: Block Diagram of the Proposed Solution for the Expert User

image for classification to the cloud. The model located in the cloud will return the label along with some images of the same determined label. The user will be requested to provide feedback. This feedback will be later used for evaluation.

Figure 2 shows the block diagram that will be dedicated to the expert user. An expert user will be able to contribute images and labels to the model. Therefore, the expert will take pictures and through the application can submit them to the model with their respective label. The proposed system will verify the image and upon verification, the image will be uploaded to the cloud to update the data set.



(a) Model classifying image (b) Users giving their feedback

Fig. 3: Implementation of the prototype application to be used in user studies

A prototype application has been set up for the user as shown in Figure 3. Figure 3a shows the first part of the application where the user will be able to upload a photo and the model will classify that photo with a level of confidence. Figure 3b shows the second part of the application where the user will be shown similar images and the user will be able to give feedback to the model. The application is making use of Google services such as Firebase⁶ and the ML Kit⁷. The application is currently locally based and is able to classify

⁶Google's cloud mobile back-end platform, available at <https://firebase.google.com/>

⁷Machine Learning for mobile development by Google, available at <https://firebase.google.com/products/ml-kit>

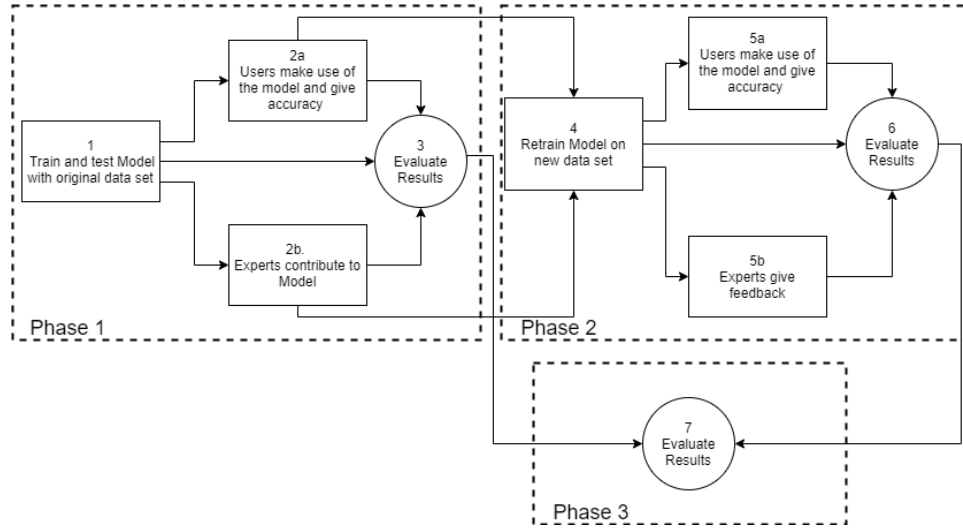


Fig. 4: Evaluation Process

between five different types of flowers: roses, sunflowers, dandelions, tulips and daisies. Upon submission of a photo either by selecting a picture from the gallery or from taking a picture with the phone camera, the model would classify the image and return a label with a level of confidence. Three images from that same label will be loaded and the user will be able to compare the flower that needed classification with the generated images. The user is then requested through the use of the rating stars to submit a rating of how accurate the model was.

VI. EVALUATION PLAN

The evaluation plan for this system is going to be performed in three phases as shown in Figure 4.

In the first phase of evaluation, the model is going to be trained on the original dataset and it is going to be tested on a test set. The users and the experts are going to contribute to the model by giving their feedback on the results generated by the model. The experts will be able to contribute new images for the dataset alongside with their respective labels. An accuracy based on the user's feedback and an accuracy based on the experts feedback will be computed. At the end of the first phase, an evaluation will be conducted and the ML model, the user's feedback and the expert's feedback will be compared as shown in Table 5.

The second phase of the evaluation involves the retraining of the model based on the new information contributed to the system. The new model with the new dataset will be tested on a test set. The users and the experts will once again contribute to the model giving their feedback on the results generate by the model. An evaluation will be conducted to compare the performance of the new model based on the users and experts feedback as shown in Table 5.

The third and final phase would conclude this study by comparing the model before and after retraining. In this phase, we would analyse if citizen science helped in the

TABLE V: Comparison of the Machine Learning Model, the user's feedback and the expert's feedback

	Expert	User	Machine Learning
Accuracy			

development of the model. Furthermore, comparisons between the performance of the user, the expert and the ML algorithm will be performed and evaluated.

In addition, focussed analysis and discussions will be carried out with some users to provide their feedback on citizen science and gamification.

The mean average precision [44] is going to be used to calculate the performance of the models. The mean average precision (mAP) is defined as:

$$\frac{\sum_{q=1}^Q AveP(q)}{Q} \quad (1)$$

where Q is the number of queries in the set and AveP(q) is the average precision for a given query q

VII. CONCLUSION

This report presents the research that has been conducted so far. Throughout this report, similar work has been investigated and a proposed solution and an evaluation plan has been designed to tackle the problem at hand; to recognize Maltese flora given the lack of data available through the use of gamification and deep learning models. In addition, an android application that was designed and developed for data collection was presented. In the coming weeks, this application will be developed further to better fit the criteria of this research and the training on the first data set will commence.

REFERENCES

- [1] S. Ray, "A quick review of machine learning algorithms," in *2019 International Conference on Machine Learning, Big Data, Cloud and Parallel Computing (COMITCon)*, pp. 35–39, Feb 2019.
- [2] K. Reynolds, A. Kontostathis, and L. Edwards, "Using machine learning to detect cyberbullying," in *2011 10th International Conference on Machine Learning and Applications and Workshops*, vol. 2, pp. 241–244, Dec 2011.
- [3] P. Shimpi, S. Shah, M. Shroff, and A. Godbole, "A machine learning approach for the classification of cardiac arrhythmia," in *2017 International Conference on Computing Methodologies and Communication (ICCMC)*, pp. 603–607, July 2017.
- [4] S. Loussaief and A. Abdelkrim, "Machine learning framework for image classification," in *2016 7th International Conference on Sciences of Electronics, Technologies of Information and Telecommunications (SETIT)*, pp. 58–61, Dec 2016.
- [5] G. A. Sandag, N. E. Tedry, and S. Lolong, "Classification of lower back pain using k-nearest neighbor algorithm," in *2018 6th International Conference on Cyber and IT Service Management (CITSM)*, pp. 1–5, Aug 2018.
- [6] M. Rizwan and D. V. Anderson, "Using k-nearest neighbor and speaker ranking for phoneme prediction," in *2014 13th International Conference on Machine Learning and Applications*, pp. 383–387, Dec 2014.
- [7] U. Aprilliani and Z. Rustam, "Osteoarthritis disease prediction based on random forest," in *2018 International Conference on Advanced Computer Science and Information Systems (ICACSIS)*, pp. 237–240, Oct 2018.
- [8] M. S. Kumar, V. Soundarya, S. Kavitha, E. S. Keerthika, and E. Aswini, "Credit card fraud detection using random forest algorithm," in *2019 3rd International Conference on Computing and Communications Technologies (IC-CCT)*, pp. 149–153, Feb 2019.
- [9] Y. Guo, D. Niu, and Y. Chen, "Support vector machine model in electricity load forecasting," in *2006 International Conference on Machine Learning and Cybernetics*, pp. 2892–2896, Aug 2006.
- [10] Jian-Wei Li, You-Hua Wang, Gui-Long Zong, and Qing Wu, "Support vector machine method using in eeg signals study of epileptic spike," in *2009 International Conference on Machine Learning and Cybernetics*, vol. 2, pp. 1241–1245, July 2009.
- [11] E. Khosla and D. Ramesh, "Phase classification of chronic myeloid leukemia using convolution neural networks," in *2018 4th International Conference on Recent Advances in Information Technology (RAIT)*, pp. 1–6, March 2018.
- [12] E. Lanfranco and G. Lanfranco, *Il-Flora Maltija*. March 2003.
- [13] G. Lanfranco, *Fjuri Slavaġ Maltin*. 2007.
- [14] A. Tóth and S. Tóvölgyi, "The introduction of gamification: A review paper about the applied gamification in the smartphone applications," in *2016 7th IEEE International Conference on Cognitive Infocommunications (CogInfoCom)*, pp. 000213–000218, Oct 2016.
- [15] N. P. Cechetti, D. Biduki, and A. C. B. De Marchi, "Gamification strategies for mobile device applications: A systematic review," in *2017 12th Iberian Conference on Information Systems and Technologies (CISTI)*, pp. 1–7, June 2017.
- [16] Y. Liu, F. Tang, D. Zhou, Y. Meng, and W. Dong, "Flower classification via convolutional neural network," in *2016 IEEE International Conference on Functional-Structural Plant Growth Modeling, Simulation, Visualization and Applications (FSPMA)*, pp. 110–116, Nov 2016.
- [17] Y. Pei and W. Cao, "A method for regional feature extraction of flower images," in *2010 International Conference on Intelligent Control and Information Processing*, pp. 384–388, Aug 2010.
- [18] Y. Li, J. Zhang, P. Gao, L. Jiang, and M. Chen, "Grab cut image segmentation based on image region," in *2018 IEEE 3rd International Conference on Image, Vision and Computing (ICIVC)*, pp. 311–315, June 2018.
- [19] T. Tiay, P. Benyaphaichit, and P. Riyamongkol, "Flower recognition system based on image processing," in *2014 Third ICT International Student Project Conference (ICT-ISPC)*, pp. 99–102, March 2014.
- [20] Y. Li, Y. Du, B. Liu, H. Li, Q. Zhang, and Y. Li, "Image segmentation method based on improved graph cut algorithm," in *2012 IEEE International Conference on Computer Science and Automation Engineering (CSAE)*, vol. 2, pp. 336–340, May 2012.
- [21] H. M. Zawbaa, M. Abbass, S. H. Basha, M. Hazman, and A. E. Hassenian, "An automatic flower classification approach using machine learning algorithms," in *2014 International Conference on Advances in Computing, Communications and Informatics (ICACCI)*, pp. 895–901, Sep. 2014.
- [22] J. Yousefi, "Image binarization using otsu thresholding algorithm," 05 2015.
- [23] E. Karami, S. Prasad, and M. S. Shehata, "Image matching using sift, surf, BRIEF and ORB: performance comparison for distorted images," *CoRR*, vol. abs/1710.02726, 2017.
- [24] M. J. Hasan, J. Uddin, and S. N. Pinku, "A novel modified sfta approach for feature extraction," in *2016 3rd International Conference on Electrical Engineering and Information Communication Technology (ICEEICT)*, pp. 1–5, Sep. 2016.
- [25] S. Hong and L. Choi, "Automatic recognition of flowers through color and edge based contour detection," in *2012 3rd International Conference on Image Processing Theory, Tools and Applications (IPTA)*, pp. 141–146, Oct 2012.
- [26] X. Xia, C. Xu, and B. Nan, "Inception-v3 for flower classification," in *2017 2nd International Conference on Image, Vision and Computing (ICIVC)*, pp. 783–787, June 2017.

- [27] N. R. Gavai, Y. A. Jakhade, S. A. Tribhuvan, and R. Bhattad, "Mobilenets for flower classification using tensorflow," in *2017 International Conference on Big Data, IoT and Data Science (BID)*, pp. 154–158, Dec 2017.
- [28] Y. Wu, X. Qin, Y. Pan, and C. Yuan, "Convolution neural network based transfer learning for classification of flowers," in *2018 IEEE 3rd International Conference on Signal and Image Processing (ICSIP)*, pp. 562–566, July 2018.
- [29] A. G. Jivani, "The novel k nearest neighbor algorithm," in *2013 International Conference on Computer Communication and Informatics*, pp. 1–4, Jan 2013.
- [30] Zhang Jian and Zhang Wei, "Support vector machine for recognition of cucumber leaf diseases," in *2010 2nd International Conference on Advanced Computer Control*, vol. 5, pp. 264–266, March 2010.
- [31] S. Na, L. Xumin, and G. Yong, "Research on k-means clustering algorithm: An improved k-means clustering algorithm," in *2010 Third International Symposium on Intelligent Information Technology and Security Informatics*, pp. 63–67, April 2010.
- [32] F. S. Mohamad, A. A. Manaf, and S. Chuprat, "Histogram matching for color detection: A preliminary study," in *2010 International Symposium on Information Technology*, vol. 3, pp. 1679–1684, June 2010.
- [33] S. Vishnupriya and K. Meenakshi, "Automatic music genre classification using convolution neural network," in *2018 International Conference on Computer Communication and Informatics (ICCCI)*, pp. 1–4, Jan 2018.
- [34] M. Han, J. Chen, L. Li, and Y. Chang, "Visual hand gesture recognition with convolution neural network," in *2016 17th IEEE/ACIS International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing (SNPD)*, pp. 287–291, May 2016.
- [35] L. Shao, F. Zhu, and X. Li, "Transfer learning for visual categorization: A survey," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 26, pp. 1019–1034, May 2015.
- [36] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Transactions on Knowledge and Data Engineering*, vol. 22, pp. 1345–1359, Oct 2010.
- [37] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 2818–2826, June 2016.
- [38] A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, and H. Adam, "Mobilenets: Efficient convolutional neural networks for mobile vision applications," 2017.
- [39] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014.
- [40] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, June 2016.
- [41] M. E. Nilsback and A. Zisserman, "17 category flower dataset." <http://www.robots.ox.ac.uk/~vgg/data/flowers/17/>. [Online]. Accessed on: November 18, 2019.
- [42] M. E. Nilsback and A. Zisserman, "102 category flower dataset." <http://www.robots.ox.ac.uk/~vgg/data/flowers/102/>. [Online]. Accessed on: November 18, 2019.
- [43] Gverzea, "Edible wild plants." <https://www.kaggle.com/gverzea/edible-wild-plants>. [Online]. Accessed on: November 18, 2019.
- [44] S. M. Beitzel, E. C. Jensen, and O. Frieder, *MAP*, pp. 1691–1692. Boston, MA: Springer US, 2009.

Work Plan

